Shyam Harimohan Menon

Mount Stromlo Observatory
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Education & Research Experience

Oct. 2019 – present
Doctor of Philosophy, *Astronomy & Astrophysics*,
Research School of Astronomy & Astrophysics, Australian National University, Canberra, Australia
Thesis title: Impacts of Stellar Feedback on Star Formation
Advisors: A/Prof Christoph Federrath & Prof. Mark Krumholz

Aug. 2014 – May 2019
Integrated Bachelor and Master of Technology (Hons.), *Engineering Physics*,
Department of Physics, Indian Institute of Technology (BHU), Varanasi, India
Thesis title: HI Fragmentation in the Turbulent Interstellar Medium
Advisor: Dr. Prasun Dutta

May 2018 – August 2019
Summer Research Scholar,
University of Tübingen, Tübingen, Germany
Project title: Impact of ionising radiation on the turbulent ISM
Advisor: Dr. Rolf Kuiper

May 2017 – August 2017
MITACS Globalink Summer Internship,
University of Western Ontario, London, Canada
Project title: Numerical Magneto-hydrodynamic simulations with the Athena code
Advisor: Prof. Shantanu Basu

Research Projects

August 2020 – present
Impact of Reprocessed Infrared Radiation Pressure on Super-Star Cluster (SSC) Formation,
Primary Collaborators:A/Prof Christoph Federrath & Prof. Mark Krumholz
• Developed `VETTAM`[1], a novel radiation-hydrodynamics module with Adaptive-Mesh Refinement in the `FLASH` code, to study the effects of radiation on star cluster formation. [Publication in prep]
• Currently running high-resolution radiation-hydrodynamic simulations of SSC formation using the `VETTAM` module to test the impact of infrared radiation pressure from massive stars on the natal molecular cloud to uncover clues on globular cluster formation

[1] Variable Eddington Tensor closed Transport on Adaptive Meshes (VETTAM)

August 2019 – May 2021 **Impact of ionising radiation feedback on the turbulent interstellar medium (ISM)**,
Primary Collaborators: A/Prof. Christoph Federrath, Dr. Rolf Kuiper, Dr. Pamela Klaassen, Piyush Sharda
• Probed the effects of ionising radiation on driving compressive turbulence and potentially triggering star formation in the ISM using numerical simulations. [Publication # 4]
• Tested the hypothesis of the above study observationally by quantifying the bulk properties of the turbulence in the Carina nebula. [Publication # 3]
• Extended the methodology of the previous study to quantify the dynamical effects of physical mechanisms on turbulence in the LMC. [Publication # 1]

September 2020 – Present **The Spatial Distribution of Star Clusters in External Galaxies**,
Primary Collaborators: Dr. Kathryn Grasha, Prof. Bruce G. Elmegreen, Prof. Daniela Calzetti, Dr. Angela Adamo
• Investigated the fractal distribution of young star clusters in 12 local galaxies with the LEGUS survey. [Publication # 2]
• Built scientific collaborations within the LEGUS survey .
• Currently extending work to probe the star cluster distributions in local high-redshift analogue galaxies with the HiPEEC survey.

Honors & Grants

Oct 2021 ANU Olin J Eggen Research Award for excellence in research

Oct 2019 ANU PhD Scholarship

Oct 2019 ANU RSAA Supplementary Research Scholarship

August 2018 Graduate Aptitude Test in Engineering (GATE) Postgraduate Scholarship

May 2018 Eberhard Karls Universität Tübingen Summer Research Stipend

May 2017 Mitacs Globalink Research Internship Award

April 2013 Kishore Vaigyanik Protsahan Yojana (KVPY) Scholarship

Relevant Skills

Languages:, English, Malayalam, Tamil, Hindi, Sanskrit

Programming Experience:, C, C++, Fortran, Python, Mathematica, Bash

Codes/Softwares:, `FLASH`, `PLUTO`, `PETSc`, `Hypre`, `Athena`, `yt`, `CUDA`

Synergistic Activities

July 2021 – present **Reviewer, Monthly Notices of the Royal Astronomical Society (MNRAS)**,
Total Reviewed Papers: 1

Sept 2020 – present **Member of Seminars Commitee, RSAA**,
Hosted 20 external colloquia and seminars at RSAA

March 2021 – present **Mentor, McNamara-Saunders Astronomical Teaching Telescopes Project**,
Mentored a grade 10 high school student on a project to measure the period of the l-Carinae Cepheid-variable star.

Jan 2020–present **Member, Astronomical Society of Australia (ASA)**,
Active student member of the ASA, and its theoretical astrophysics working group ANITA.

Jan 2020–present **Student Representative, Work, Health & Safety Committee, RSAA**,
Formally representing the student cohort in ensuring a safe and healthy workplace.

Sep 2021 – present **Standard Mental Health First Aider (MHFA)**,
Accredited MHFA by Mental Health First Aid Australia

Aug 2017 – Nov 2017 **Organising Core Team Member, Jigyasa 2017**,
Core team member in organising Jigyasa, the annual physics convention held at the Department of Physics, IIT(BHU) Varanasi

Publications

1. First extragalactic measurement of the turbulence driving parameter: ALMA observations of the star-forming region N159E in the Large Magellanic Cloud
Sharda, P., **Menon, S.H.**, Federrath, C., Krumholz, M. R., Beattie, J. R., Jameson, K. E., Tokuda, K., Burkhart, B., Crocker, R. M., Law, C. J., Seta, A., Gaetz, T. J., Pingel, N. M., Seitenzahl, I. R., Sano, H., and Fukui, Y., MNRAS, accepted (arXiv: 2109.03983)
2. The dependence of the hierarchical distribution of star clusters on galactic environment
Menon, S.H., Grasha, K., Elmegreen, B.G., Federrath, C., Krumholz, M.R., Calzetti, D., Sánchez, N., Linden, S.T., Adamo, A., Messa, M., Cook, D.O., Dale, D.A., Grebel, E.K., Fumagalli, M., Sabbi, E., Johnson, K.E., Smith, L.J., Kennicutt, R.C., 2021, MNRAS, 507, 5542
3. On the compressive nature of turbulence driven by ionizing feedback in the pillars of the Carina Nebula [7 Citations]
Menon, S.H., Federrath, C., Klaassen, P., Kuiper, R., Reiter, M., 2021, MNRAS, 500, 1721
4. On the turbulence driving mode of expanding H II regions [9 Citations]
Menon, S.H., Federrath, C., Kuiper, R., 2020, MNRAS, 493, 4643

Conference Presentations & Colloquia

Nov 2021 **Contributed Talk, IAU Symposium 362 - The Predictive Power of Computational Astrophysics**,
Title: VETTAM - A novel algorithm for modelling radiation hydrodynamics

May 2021 **Contributed Talk, ISM 2021, Beirut - Structure, Characteristic Scales, and Star Formation**,
Title: Spatial distribution of star clusters in external galaxies

February 2021 **Contributed Talk, Australian National Insitute for Theoretical Astrophysics science workshop**,
Title: Compressive Turbulence Driven by Ionising Radiation in the ISM

September 2020 **Contributed Talk, Annual meeting of the Astronomische Gesellschaft, Germany**,
Title: Compressive Turbulence Driven by Ionising Radiation

September 2020 **Contributed Talk, Annual meeting of the Astronomische Gesellschaft, Germany**,
Title: Compressive Turbulence Driven by Ionising Radiation

March 2020 **Invited Talk, Modelling High-Mass Stellar Feedback Workshop, Tübingen, Germany**,
Title: Turbulence and Triggered Star Formation in Star Clusters

January 2020 **Australian National Insitute for Theoretical Astrophysics science workshop**,
Title: Turbulence Driving Mode of Expanding HII regions

November 2019 **Mount Stromlo Student Seminars**,
Title: On the turbulence driving mode of expanding HII regions